

Muhammad Abdullah Khan

Software Engineer | Applied AI & Mobile Systems

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SUMMARY

Software Engineer specializing in applied Machine Learning, edge-optimized system architecture, and AI-powered mobile applications. Expert in deploying hardware-constrained local LLMs, building autonomous GUI agents, and developing high-performance parallel computing solutions. Demonstrated ability to translate complex AI research into production-grade products, leveraging Flutter, Docker, and scalable database architectures to build highly efficient, intelligent software systems.

EDUCATION

COMSATS University Islamabad <i>Bachelor of Science in Software Engineering (BSSE)</i>	09/2023 – Present Lahore, Pakistan
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SKILLS

AI & Machine Learning Hardware-Constrained Fine-Tuning (PEFT, LoRA, Unsloth), Model Quantization & KV Cache Optimization, Local LLM Deployment, Retrieval-Augmented Generation (RAG), ChromaDB, Episodic Memory, Set-of-Mark (SOM) Prompting, UI Tree Parsing, Pattern Recognition.	Mobile Development Flutter, Dart, Riverpod, Custom Database Backends, Firebase/Firestore, Premium UI/UX Design, Cross-Platform Architecture.	High-Performance Computing Parallel algorithm design, GPU acceleration (CUDA), distributed computing architectures (MPI), thread-level concurrency (OpenMP), and hardware-aware performance optimization.
Backend & DevOps PostgreSQL, Docker/Dockerization, Railway Cloud Hosting, Firebase/Supabase, RESTful APIs, CI/CD.	Software Engineering System Architecture, Requirements Analysis, Legacy Re-engineering, Code Optimization & Refactoring, Quality Assurance, Full-Cycle SDLC.	Web Development & Backend Full-Stack Web Architecture, CRM Development, Lead Generation Interfaces, PostgreSQL, RESTful APIs.

PROJECTS

OmniVLA <i>Hardware-Constrained Autonomous GUI Agent</i> Engineered a hardware-constrained, localized LLM GUI agent optimized via KV cache quantization to execute natively within a <6GB VRAM environment.	01/2026 – Present
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- Fine-tuned the LLM backbone for precise coordinate grounding, enabling the agent to map semantic intent to exact pixel-level screen coordinates with high spatial accuracy.
- Developed a novel semantic parsing pipeline combining Set-of-Mark (SOM) prompting, automated UI tree extraction, and localized computer vision for high-fidelity screen comprehension.
- Integrated a Retrieval-Augmented Generation (RAG) architecture using ChromaDB to establish episodic memory, enabling persistent, long-term contextual recall across extended user sessions.

QuantPDC

09/2025 – 12/2025

Distributed Algorithmic Backtesting Engine

- Engineered a high-performance, distributed algorithmic trading backtester leveraging parallel computing architectures (CUDA, MPI, OpenMP) to massively accelerate the evaluation of complex financial models.
- Architected a dynamic execution engine allowing for the seamless ingestion of custom historical datasets and the rapid integration of user-defined trading strategies.
- Implemented highly optimized, thread-safe profit and loss (PnL) calculation pipelines to handle vast arrays of tick data without memory bottlenecks.
- Developed a custom telemetry and visualization module to render granular performance graphs, real-time financial analytics, and drawdown metrics.

Stride

01/2025 – 12/2025

AI-Driven Physiological Tracking System

- Developed a production-grade fitness application utilizing Flutter and Riverpod, prioritizing efficient cross-platform performance and a premium, responsive UI/UX architecture.
- Designed a hybrid intelligence system, combining custom-trained machine learning models with the Gemini API to dynamically generate and serve personalized physiological routines.
- Architected a scalable serverless backend leveraging Firebase/Firestore, paired with Dockerized ML inference pipelines for reliable, low-latency model serving.

Sous

05/2025 – 08/2025

Intelligent Resource Recommendation Engine

- Built a cross-platform consumer recommendation engine in Flutter, integrating an interactive frontend featuring a fictional chef persona driven by ingredient-based sorting logic.
- Engineered an end-to-end data pipeline, from preprocessing large-scale Kaggle datasets to training robust ML models for real-time pattern recognition.
- Containerized the complete machine learning architecture using Docker, establishing a decoupled, scalable microservice that integrates seamlessly with the mobile client via REST APIs.